Paper DSCI 445 Project

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1 Motivation

Mental health among university students represents a critical public health concern globally. With increasing academic, social, and financial pressures, students face a significant risk of developing mental health issues, with depression being one of the most prevalent conditions. Early and accurate detection of at-risk students is vital for providing timely intervention, but traditional screening methods can be resource-intensive and often miss subclinical cases.

1.1 Research Question and Data

This study addresses the need for scalable and accurate early detection tools by employing machine learning classification models to predict the presence of depression in a student population. We utilized a publicly available Kaggle dataset that captures various student life factors, including Academic Pressure, Cumulative GPA (CGPA), Study Satisfaction, Sleep Duration, and Financial Stress. The goal is to determine which of these common student experience factors are the most significant predictors of depression.

1.2 Project's Objective

The primary objective of this project is to compare the predictive accuracy and interpretability of four distinct machine learning algorithms, Logistic Regression, LASSO-Regularized Logistic Regression, K-Nearest Neighbors (KNN), and Random Forest (RF), on this binary classification task. By benchmarking models ranging from simple linear approaches (Logistic Regression) to complex ensemble methods (Random Forest), we aim to identify the most robust, accurate, and practically deployable model. The final selected model must not only demonstrate high performance on unseen data but also provide actionable, interpretable insights to inform mental health support programs.

2 Methodology

2.1 Data Preparation and Cleaning

The project began with standardized data preparation. Initial cleaning involved removing irrelevant or redundant columns (e.g., Student ID, Work Pressure, Job Pressure) which contained no variance or excessive missing values. The target variable, Depression (0/1), was confirmed to be a binary factor. To manage high-cardinality nominal predictors:

Academic Level: The various degree programs were consolidated into four primary categories: High school, Undergraduate, Masters, and PhD. City: The City variable was processed using the fct_lump(n=10) method, consolidating cities with low frequency into a single "Other" factor level. This reduced feature instability and complexity for the linear models.

2.2 Experimental Design and Evaluation

The entire dataset was partitioned into a Training Set (80%) and a permanently Held-Out Test Set (20%) using a stratified sampling approach to ensure the target variable distribution was preserved in both sets. All model tuning and cross-validation were conducted exclusively on the Training Set, and a consistent seed of set.seed(445) was used. Model Selection: All models were tuned using a consistent 5-fold Cross-Validation (CV) or 10-fold CV procedure on the Training Set to select the optimal hyperparameters (e.g., k for KNN, λ for LASSO). Final Evaluation: The accuracy of the final, tuned models (LASSO, KNN, RF, and Logistic Regression) was determined using the Test Set Accuracy, providing an unbiased comparison of generalization performance on unseen data.

2.3 Exploratory Data Analysis

Prior to model fitting, exploratory data analysis was conducted to examine distributions and relationships among predictors. Visualizations included histograms for numeric variables, bar plots for categorical variables, and boxplots comparing numeric predictors across depression status. These plots revealed clear patterns, including higher financial stress among students reporting suicidal thoughts and concentration of depression cases among students experiencing elevated academic pressure. These findings motivated the inclusion of these predictors in subsequent logistic regression models.

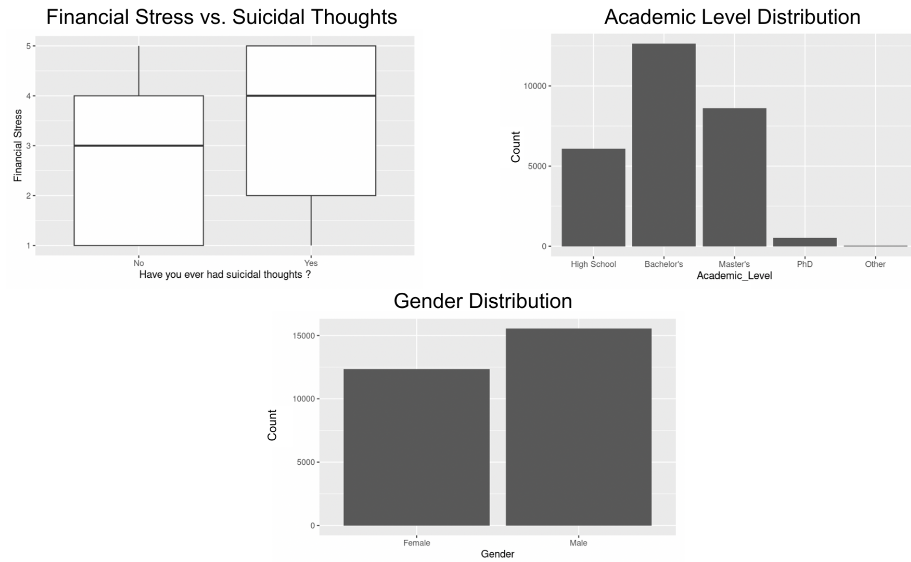


Figure 1. Example exploratory plots showing relationship between Financial Stress and Suicidal Thoughts, Academic Level distribution, and Gender distribution.

2.4 Logistic Regression (Unregularized)

A standard, unregularized Logistic Regression model was developed to serve as the baseline for the linear models. Model selection was performed by fitting 16 distinct models, incrementally adding predictors from a single feature up to the full set of 13 base predictors, followed by three models testing intuitive interaction terms (e.g., Sleep Duration * Work/Study Hours). The 5-fold CV process identified Model 13 (the full model with 13 base predictors) as having the highest average CV accuracy (0.84791). This model was selected for final Test Set evaluation.

5 Fold Cross Validation Results
for Logistic Regression Models

Model	Predictors	Accuracy
Model 1	1	0.5855258
Model 2	2	0.6324466
Model 3	3	0.6320524
Model 4	4	0.7436733
Model 5	5	0.7436375
Model 6	6	0.7498028
Model 7	7	0.7491216
Model 8	8	0.7606277
Model 9	9	0.8250411
Model 10	10	0.8300235
Model 11	11	0.8460464
Model 12	12	0.8477669
Model 13	13	0.8479103
Model 14	14	0.8476235
Model 15	15	0.8478027
Model 16	16	0.8478386

Table 1. Cross validation accuracy for 16 logistic regression models, highlighting that Model 13, with all 13 predictors and no interaction terms, achieved the highest mean accuracy.

2.5 LASSO-Regularized Logistic Regression

Prior to fitting the linear models, a data recipe was applied to ensure the model prerequisites were met. This process included Dummy Encoding of all categorical predictors (e.g., Gender, Academic Level) to convert them into a set of binary (0/1) features compatible with the Logistic Regression equation. Furthermore, all numeric predictors were normalized (scaled) to have a mean of zero and a standard deviation of one. This step was critical for the LASSO model to ensure the L1 regularization penalty was applied equally and fairly across all predictors, preventing bias in feature selection based solely on variable scale.

To enhance the performance and interpretability of the linear model, LASSO (L1 Regularization) was implemented. The LASSO model simultaneously performs shrinkage and automatic feature selection by penalizing the magnitude of the coefficients (β_j). The optimal regularization strength, λ, was determined via 5-fold cross-validation. The final model exhibited high sparsity, setting 27 of the initial features to zero, thus retaining only the most predictive variables for evaluation. The Lambda value of 0.00174 was chosen, corresponding to the λ that yielded the minimum mean cross-validated error ($\lambda_{\min}$) across the 5-fold iterations.

2.6 Random Forest (RF)

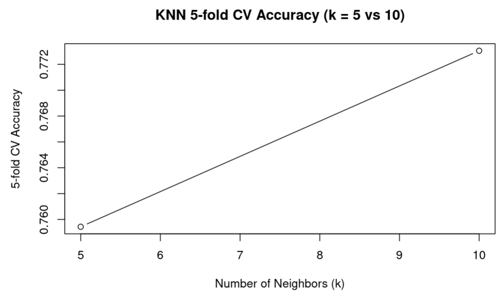
The Random Forest ensemble method was included as a non-linear classifier to capture potential complex interactions not evident in the linear models. The model was initialized with 500 decision trees to balance predictive performance against runtime. Hyperparameters, including the number of variables randomly sampled as candidates at each split (mtry), were tuned via 5-fold CV. Following tuning, the model's global Variable Importance was assessed using the Mean Decrease in Accuracy metric.

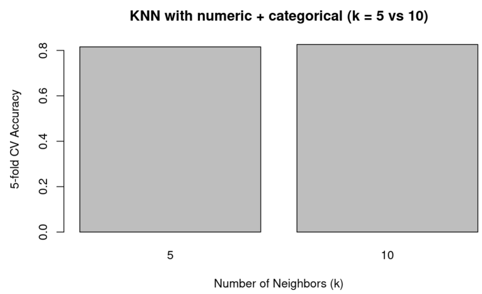
2.7 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a distance-based classifier that assigns a class label using the majority vote among the k closest training observations (Euclidean distance). Because KNN relies on distances, all predictors must be numeric and on comparable scales.

KNN was evaluated in two stages. First, a **numeric-only baseline** was fit using Age, Academic Pressure, Work Pressure, CGPA, Study Satisfaction, Job Satisfaction, Work/Study Hours, and Financial Stress. This baseline selected $k = 10$ via 5-fold cross-validation (CV accuracy = 0.7747) and achieved test accuracy = 0.7702.

For the final KNN model reported in the Results section, **numeric and categorical predictors** were used by converting categorical variables (Gender, Sleep Duration, Dietary Habits, Suicidal Thoughts, and Family History of Mental Illness) into dummy variables using `model.matrix(...)` (intercept removed). The hyperparameter $k \in \{5, 10\}$ was tuned with **5-fold cross-validation**, and **fold-specific standardization** was applied (means and standard deviations computed on the fold training subset and applied to the fold validation subset). The best model was $k = 10$ (CV accuracy = 0.8263). On the held-out test set, this model achieved **test accuracy** = 0.8271 **(82.7%)** with confusion matrix counts TN = 1699, FP = 605, FN = 360, and TP = 2917.





2.8 Key Findings

One of our key findings was that for every one standard deviation increase in Academic Pressure, the odds of a student experiencing depression increase by over 200% (a factor of 3.03).)

2.9 Test Set Accuracy

The final test set scores for each model is as follows:

- Tanner RF: 89.48%
- Juliette Lasso: 85%
- Munisa KNN: 82.7%
- Leah Logistic Regression (Model 13): 84.35%

This result is distinct from the original best model being Lasso, as previously the Random Forest Cross Validation was being run with a 70% training 30% test split. This final score now reports the test set accuracy score using the consistent (80/20) split across all models.

2.10 Discussion

The high RF score suggests that the relationship between predictors and depression does involve complex interactions (e.g., high Academic Pressure only matters when Sleep Duration is low) that the other models could not capture as well as random forest.

3 References

Adil Shamim, Student Depression Dataset, kaggle.com